

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR



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Micro Project-2 on Full-Wave Bridge Rectifier Using Four Discrete Diodes.

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Submitted to

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DECLARATION

I hereby declare that the project work entitled "**Full-Wave Bridge Rectifier Using Four Discrete Diodes.**" submitted to the Madhav Institute of Technology & Science Gwalior, is a record of an original work done by **Akshat Gupta & Amit Kumar Mishra** under the guidance of **Dr. R.P. Narwaria**, Department of Electronics Engineering.

The results embodied in this micro project-2 have not been submitted to any other University or Institute for the award of any degree or diploma.

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Date: 21/04/2026.

Place: Gwalior

CERTIFICATE

This is to certify that the Micro Project report entitled " **Full-Wave Bridge Rectifier Using Four Discrete Diodes.**" submitted by **Akshat Gupta & Amit Kumar Mishra** has been carried out under the guidance of **Dr. R.P. Narwaria**, Department of Electronics Engineering, Madhav institute of technology & Science, Gwalior. The Project report is approved for submission requirement for Micro Project in 2nd semester in Electronics Engineering, from **Madhav Institute of Technology & Science, Gwalior (M.P.)**

Submitted To:

Dr. R.P. Narwaria

Signature: _____

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OBJECTIVE

To design and construct a full-wave bridge rectifier using four discrete diodes, and to study its operation by converting an alternating current (AC) input signal into a pulsating direct current (DC) output using a function generator.

ABSTRACT

A full-wave bridge rectifier is an essential electronic circuit used to convert alternating current (AC) into direct current (DC). This project focuses on the design and implementation of a full-wave bridge rectifier using four discrete diodes. An AC input signal is applied using a function generator, and the output is observed using a cathode ray oscilloscope (CRO).

The bridge configuration allows both halves of the AC waveform to be utilized, resulting in a pulsating DC output with higher efficiency compared to a half-wave rectifier. During the experiment, the input and output waveforms are analyzed to understand the rectification process and waveform transformation. The project demonstrates how the polarity of the output remains the same for both half cycles of the input signal.

This study helps in understanding the working principle of rectifier circuits, their practical implementation on a breadboard, and their importance in electronic devices that require DC power supply.

APPARATUS REQUIRED

S.No.	Apparatus	Quantity
1.	PN junction diodes (1N4007)	4
2.	Function generator (to provide AC input signal)	1
3.	Cathode Ray Oscilloscope (CRO)	1
4.	Breadboard	1
5.	Jumper Wires (Male–Male, Male–Female)	14
6.	Resistor (10 k Ω)	1
7.	LEDs	4

THEORY

A rectifier is an electronic circuit used to convert **alternating current (AC) into direct current (DC)**. A full-wave bridge rectifier is one of the most efficient rectifier circuits, as it utilizes both the positive and negative half cycles of the AC input signal.

The full-wave bridge rectifier consists of **four PN junction diodes** connected in a bridge configuration. During the positive half cycle of the input AC signal, **two diagonally opposite diodes become forward biased and conduct current**, while **the other two diodes are reverse biased and block current**. During the negative half cycle, the roles of the diodes are reversed, but the direction of current through the load resistor remains the same.

As a result, both halves of the AC input waveform are converted into a pulsating DC output. The output voltage is not constant but consists of positive pulses, which can be further smoothed using a filter.

The main advantage of a **full-wave bridge rectifier over a half-wave rectifier** is that it provides **higher efficiency**, better utilization of the input signal, and **a higher average output voltage**. It is widely used in power supply circuits of electronic devices.

WORKING PRINCIPLE

The full-wave bridge rectifier works on the principle of converting both half cycles of an AC input signal into a unidirectional (DC) output using four diodes arranged in a bridge configuration.

During the **positive half cycle** of the AC input, two diagonally opposite diodes become forward biased and conduct current, while the other two diodes are reverse biased. The current flows through the load resistor in one direction.

During the **negative half cycle**, the other pair of diodes becomes forward biased, and the previously conducting diodes become reverse biased. Although the input polarity reverses, the current still flows through the load resistor in the same direction.

Thus, in both half cycles, the current through the load remains unidirectional. This results in a pulsating DC output where both halves of the AC waveform are utilized.

1. Positive Half Cycle of Input

- ❖ The upper terminal of the AC supply becomes positive and the lower terminal becomes negative.
- ❖ Diodes D1 and D3 are **forward biased** → they **conduct**.
- ❖ Diodes D2 and D4 are **reverse biased** → they **do not conduct**.

2. Negative Half Cycle of Input

- ❖ The polarity reverses: upper terminal becomes negative, lower becomes positive.
- ❖ Diodes D2 and D4 are now forward biased → they conduct.
- ❖ Diodes D1 and D3 are reverse biased → they do not conduct.

CIRCUIT DIAGRAM

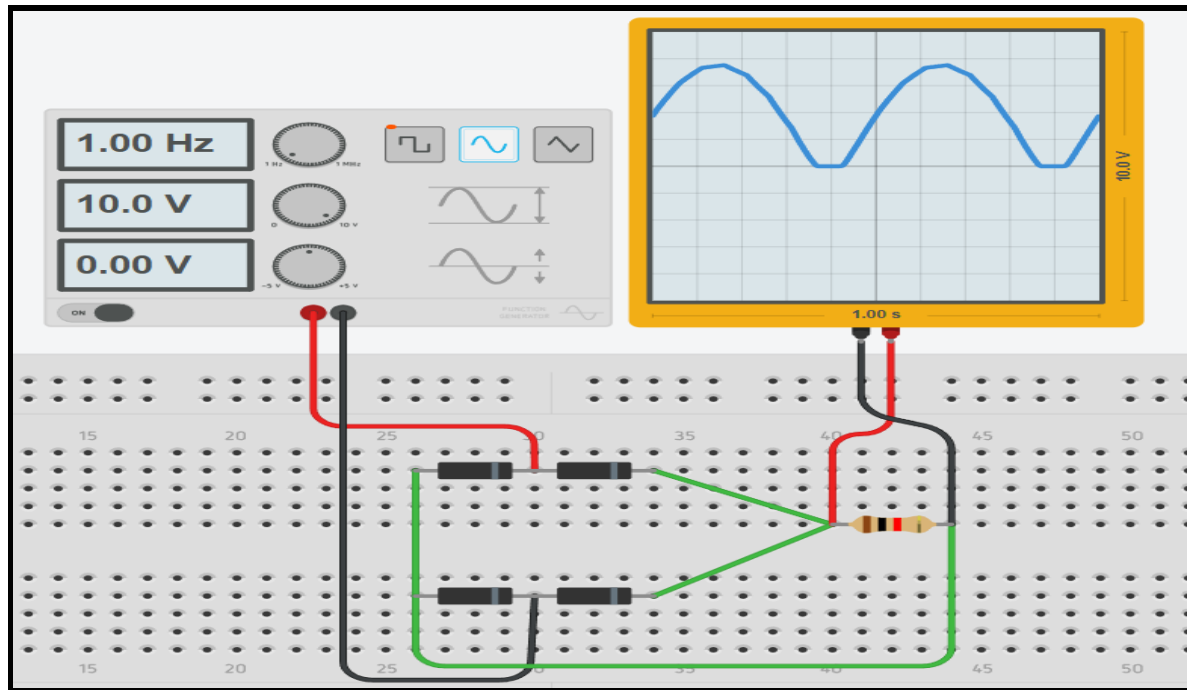


Fig.01 : Simulation of Full-Wave Bridge Rectifier on Tinkercad

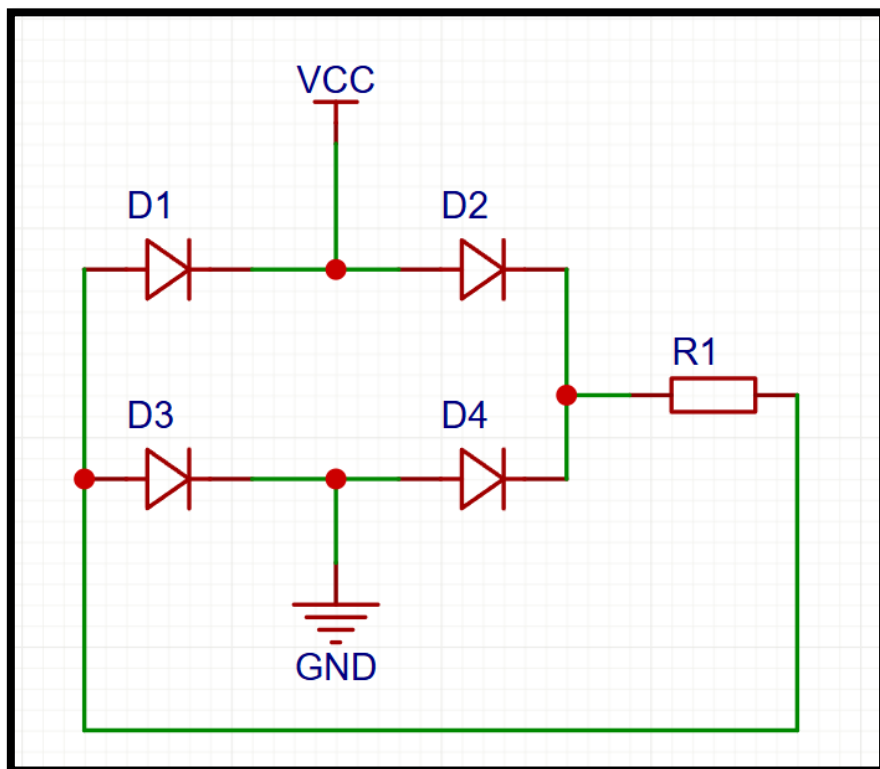


Fig.02 : Full-Wave Bridge Rectifier

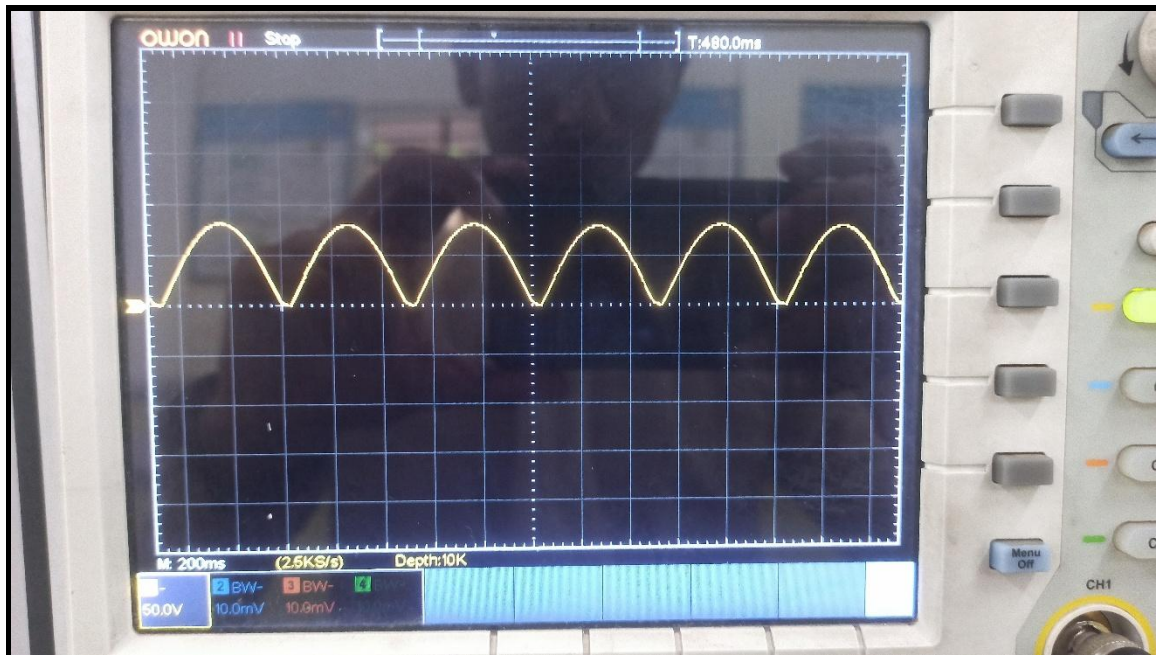


Fig.03 : Oscilloscope (CRO).

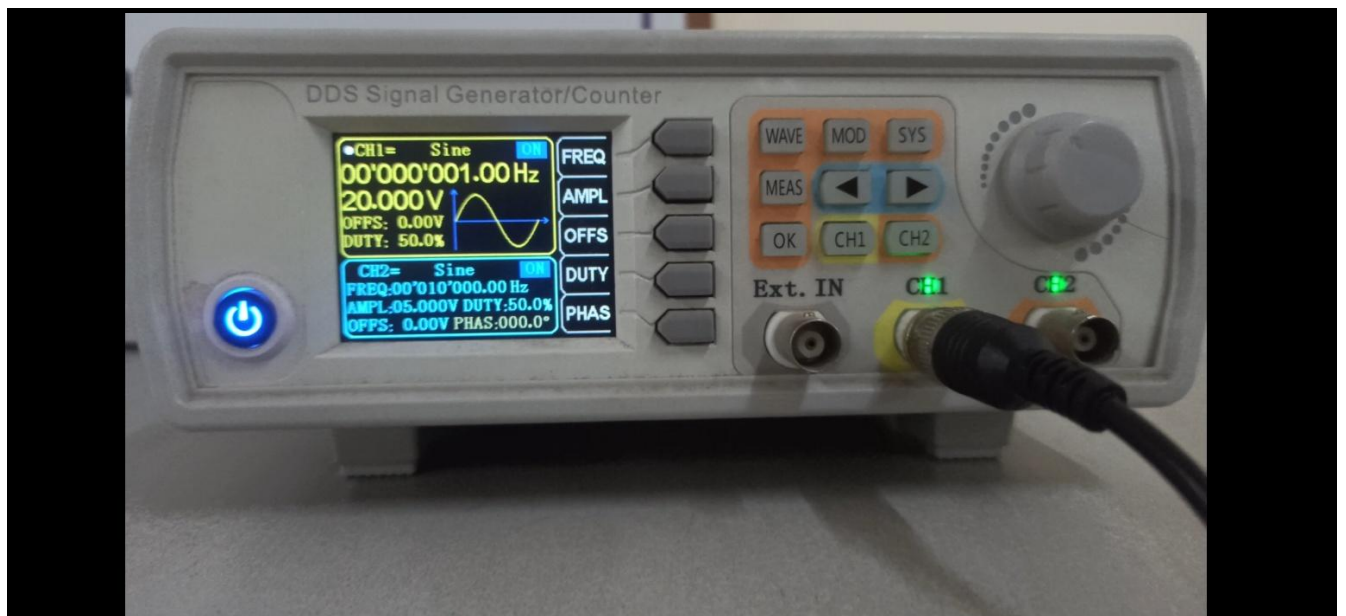


Fig.04 (a) : The Function generator .

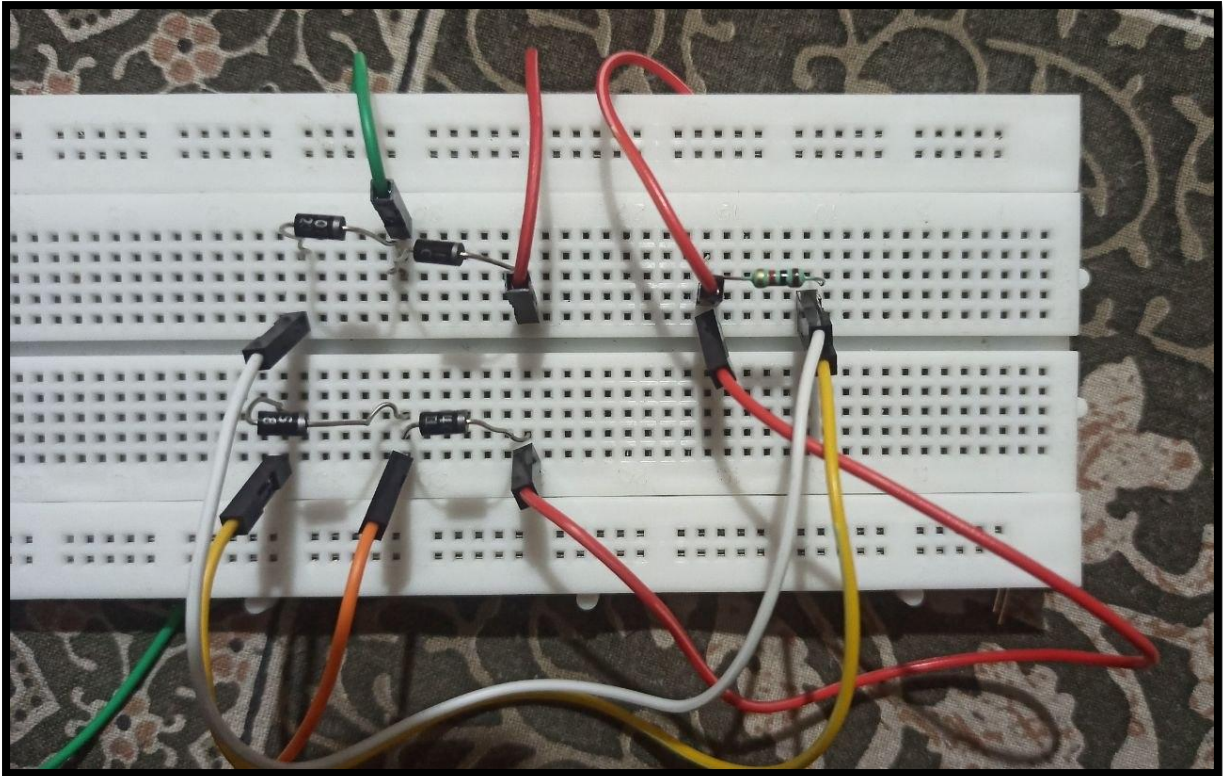


Fig.05 (a) : Full-Wave Bridge Rectifier Circuit applied on breadboard.

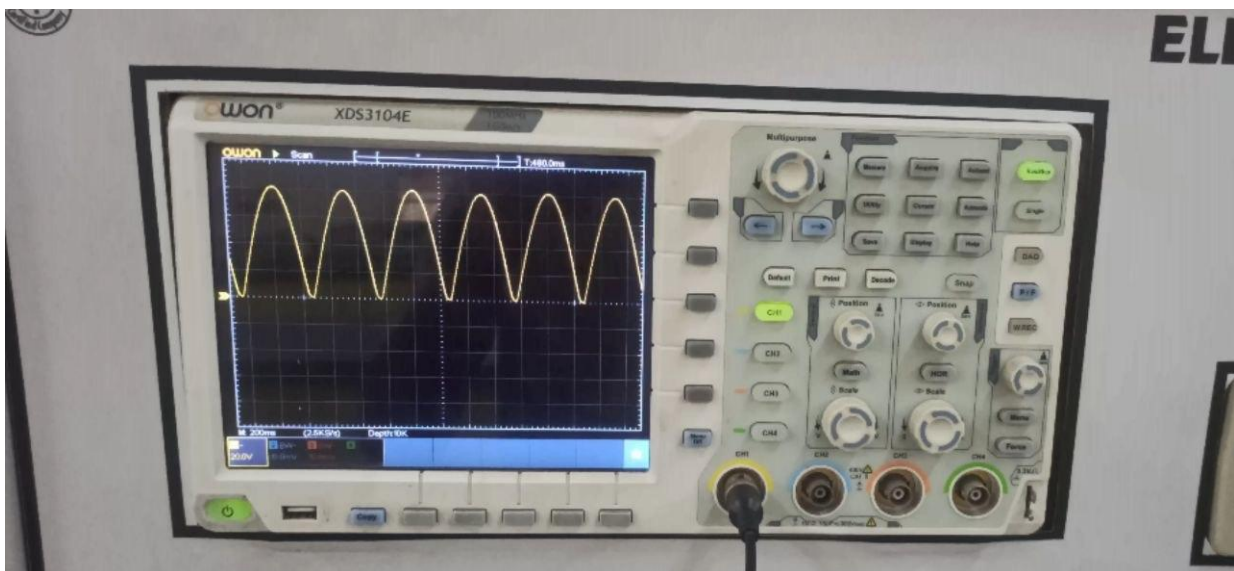


Fig.05 (b) : Successful Observed full-wave rectified output waveform on CRO showing pulsating DC with both half cycles rectified.

PCB Layout and Gerber Files Views

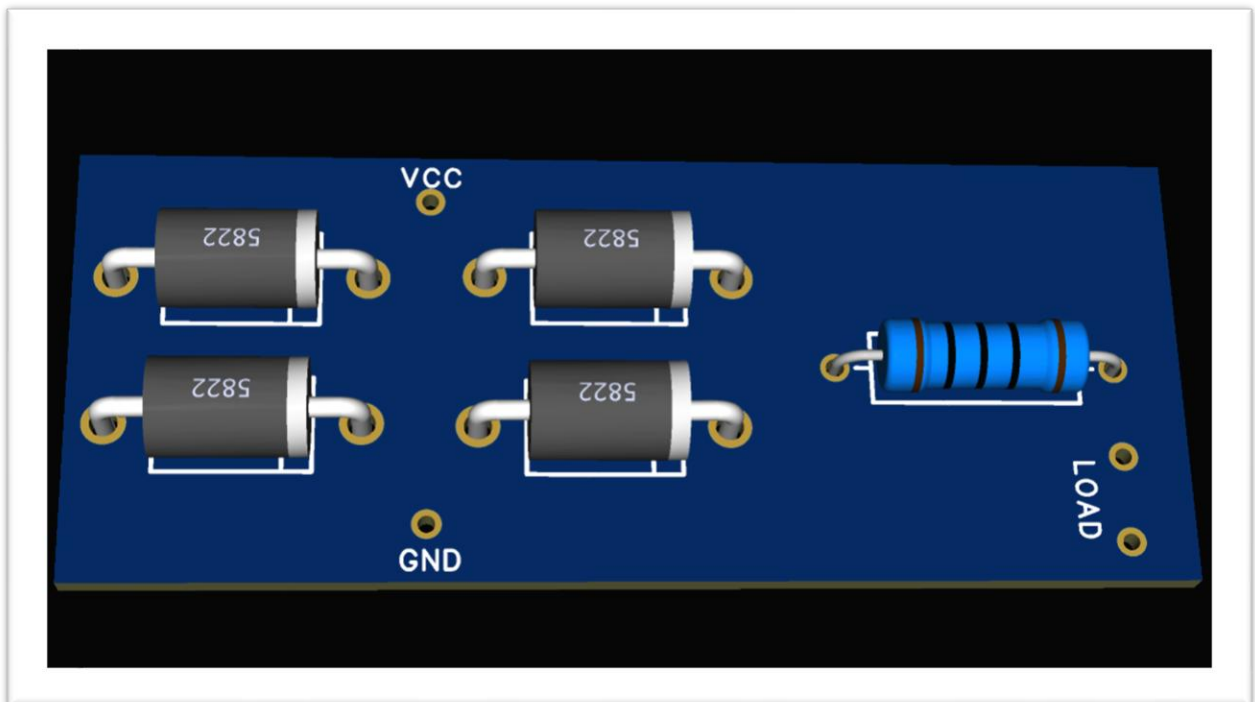


Fig. 6(a) : PCB Layout (Top View) 3-D.

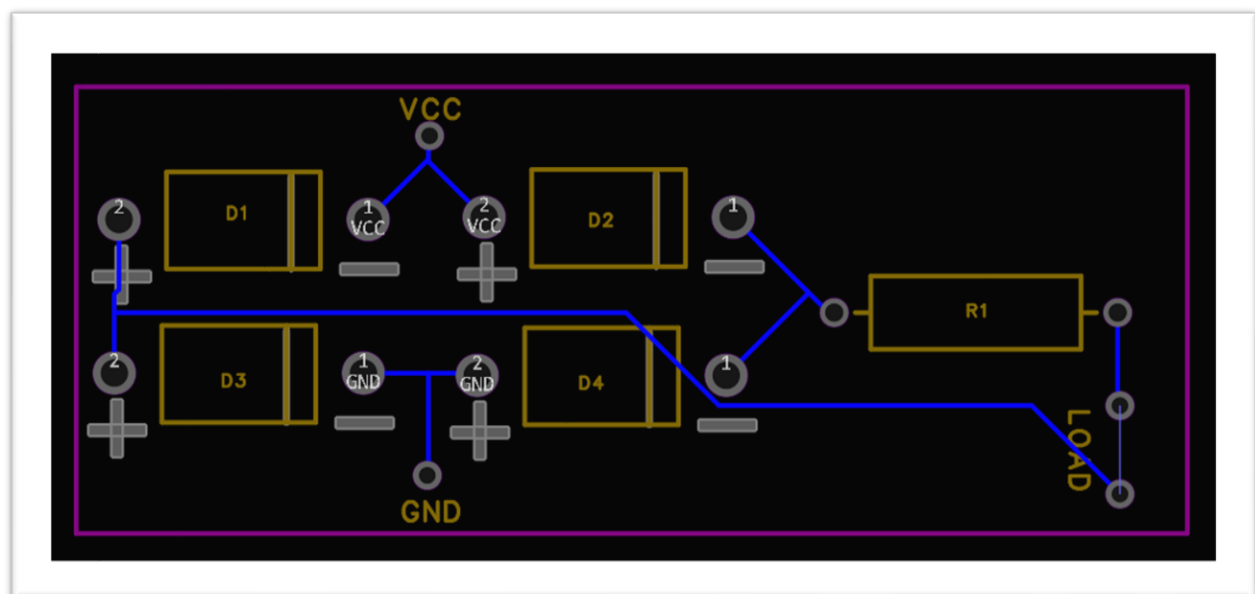


Fig. 6(b) : PCB Layout (Bottom View) 2-D.

- ❖ The Gerber files generated for PCB fabrication are provided through the link below for reference and download:

Link: [https://drive.google.com/file/d/1Dmon-](https://drive.google.com/file/d/1Dmon-kjjoJg4W8N14zOxn4fWWMRYXssR/view?usp=sharing)

[kjjoJg4W8N14zOxn4fWWMRYXssR/view?usp=sharing](https://drive.google.com/file/d/1Dmon-kjjoJg4W8N14zOxn4fWWMRYXssR/view?usp=sharing)

PROCEDURE

- ❖ Connect four PN junction diodes in a bridge configuration on the breadboard.
- ❖ Connect a load resistor (R_L) across the output terminals of the bridge rectifier.
- ❖ Connect the input terminals of the rectifier circuit to the function generator.
- ❖ Set the function generator to produce a sinusoidal AC signal.
- ❖ Connect the CRO across the input terminals to observe the input waveform.
- ❖ Connect the CRO across the load resistor to observe the output waveform.
- ❖ Switch on the function generator and apply the input signal to the circuit.
- ❖ Observe and record the input and output waveforms on the CRO.
- ❖ Verify that the output waveform is a pulsating DC signal and both half cycles are rectified.

RESULT

The full-wave bridge rectifier using four discrete diodes was successfully constructed and tested. The circuit effectively converted the applied AC input signal into a pulsating DC output. The observed output waveform on the CRO confirmed that both half cycles of the input were rectified, producing a unidirectional waveform with double the input frequency.

OBSERVATIONS

I observed that using LEDs in a full-wave bridge rectifier circuit makes the output highly sensitive. When LEDs are connected before the bridge (in the input side), the waveform gets distorted and differs from the expected sinusoidal input. When LEDs are connected after the rectifier circuit (across the load), the output waveform shows reduced amplitude due to the higher forward voltage drop of the LEDs.

I also observed that the rectifier circuit without any LEDs produces a proper full-wave rectified waveform with clear positive pulses. Hence, the presence and position of LEDs significantly affect the shape and amplitude of the output waveform.

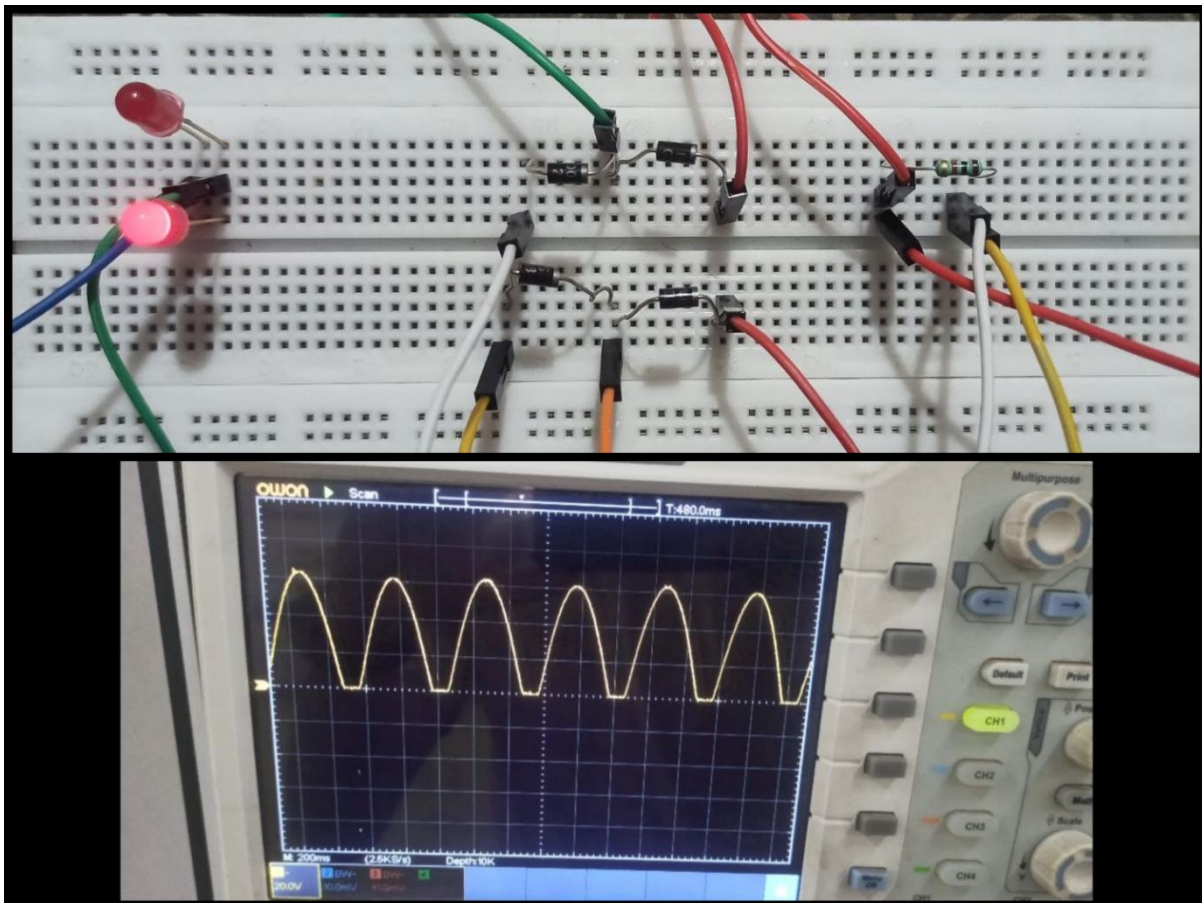


Fig. 07 (a) : When LED is at input Circuit Diagram and waveform.

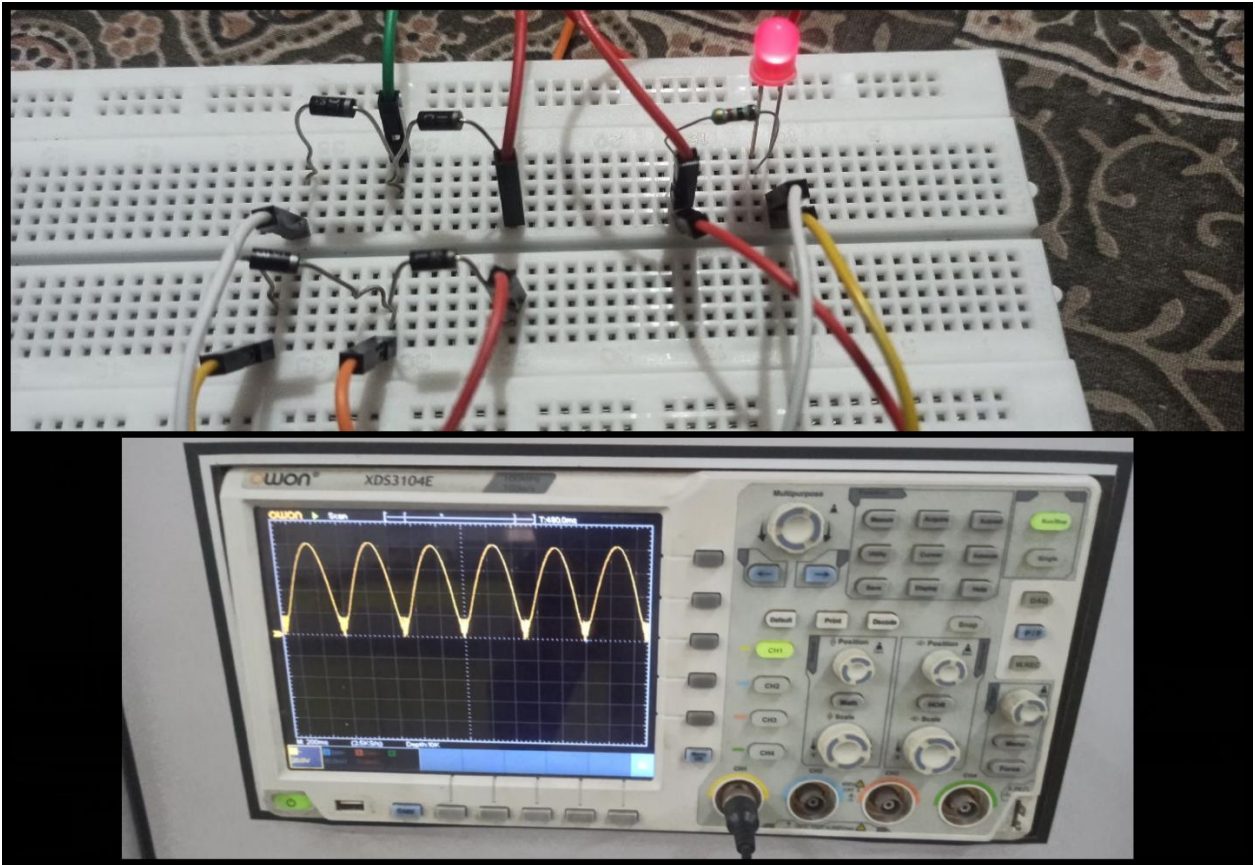


Fig. 07 (b) : When LED is at output Circuit Diagram and waveform.

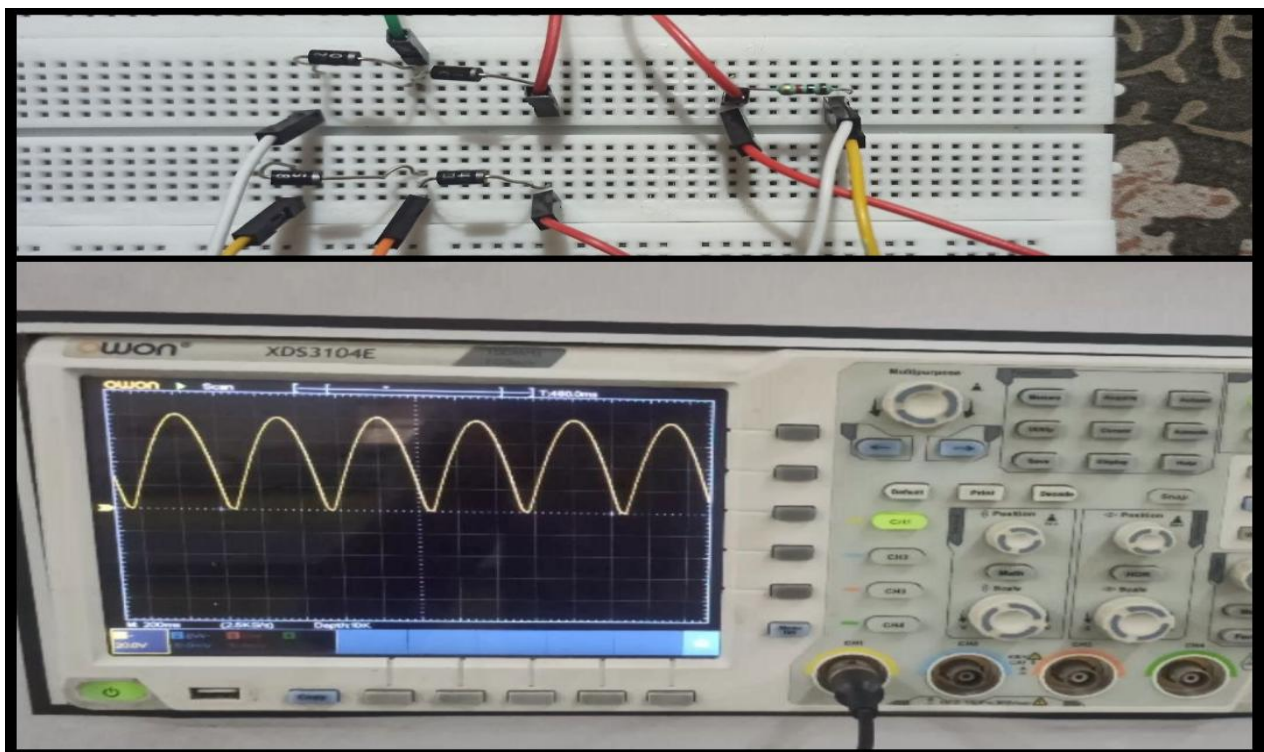


Fig. 07 (c) : Circuit Diagram and waveform with no LED

Configuration	Waveform Appearance	Why it looks this way
Without LED	Ideal Pulsating DC. Smooth, rounded humps that touch (or nearly touch) the zero-line.	The circuit is linear. Only the bridge diodes (1.4V total) must be overcome, which happens very quickly.
LED After Bridge	Notched Humps. Sharp "V" shapes or spikes appear at the bottom of each hump.	The LED turns off abruptly when $V < V_f$. The "notch" is the measurement point reacting to the LED's non-linear resistance.
LED Before Bridge	Truncated Humps. Wide, flat horizontal lines (Dead Zones) appear between each hump.	The LED acts as a gate for the entire circuit. No current reaches the bridge or load until the AC peak exceeds $V_f + 1.4V$.

APPLICATIONS

- ❖ Used in power supply circuits to **convert AC to DC**.
- ❖ Used in battery charging circuits.
- ❖ Used in electronic devices like TVs, radios, and computers.
- ❖ Used in DC power supplies for laboratory and industrial applications.
- ❖ Used in adapters and rectifier units.

CONCLUSION

The full-wave bridge rectifier using four discrete diodes was successfully designed and implemented. The circuit efficiently converted the AC input into a pulsating DC output by utilizing both half cycles of the input signal. The observed waveforms on the CRO matched the expected results, confirming proper operation of the circuit.

It was also concluded that the **use of LEDs** in the circuit **affects the output waveform** due to their higher forward voltage drop, which leads to reduced amplitude and possible distortion. Thus, proper component selection is important for accurate rectification.

REFERENCES

- ❖ Wikipedia – “Diode” and “Rectifier” articles
- ❖ Electronic Devices and Circuit Theory – Robert L. Boylestad & Louis Nashelsky
- ❖ Tinkercad – Used for circuit simulation and testing of rectifier circuit
- ❖ All About Circuits – Tutorials on rectifiers and diodes
- ❖ Electronics Tutorials – Bridge Rectifier and LED working